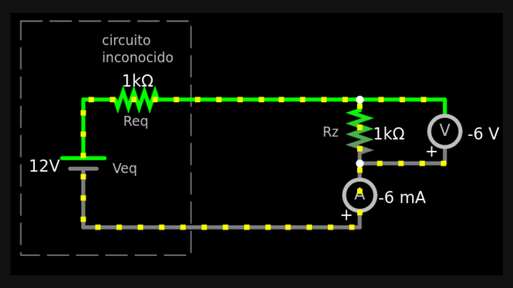
circuito 1

(1) $V = \frac{R_z}{R_z + R_{eq}} V_{eq}$

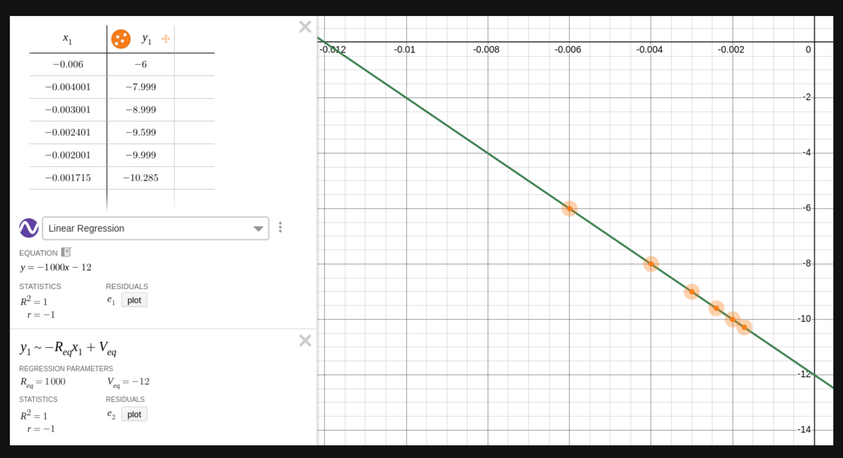
(2) $A = \frac{V_{eq}}{R_z + R_{eq}}$



usamos (2) en (1)

$$V = R_z A \quad \Rightarrow \quad R_z = \frac{V_{eq}}{A} - R_{eq}$$

$$V = -R_{eq} A + V_{eq}$$



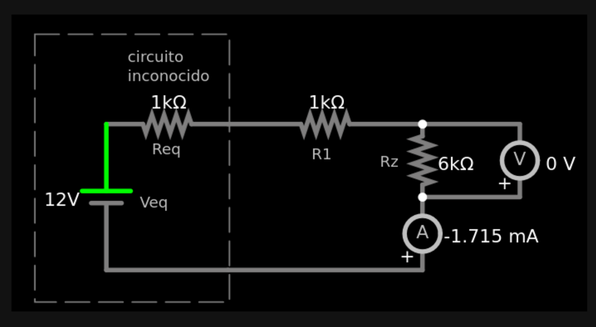
$R_{eq} = 1000\,\Omega$

$V_{eq} = -12\,V$

circuito 2

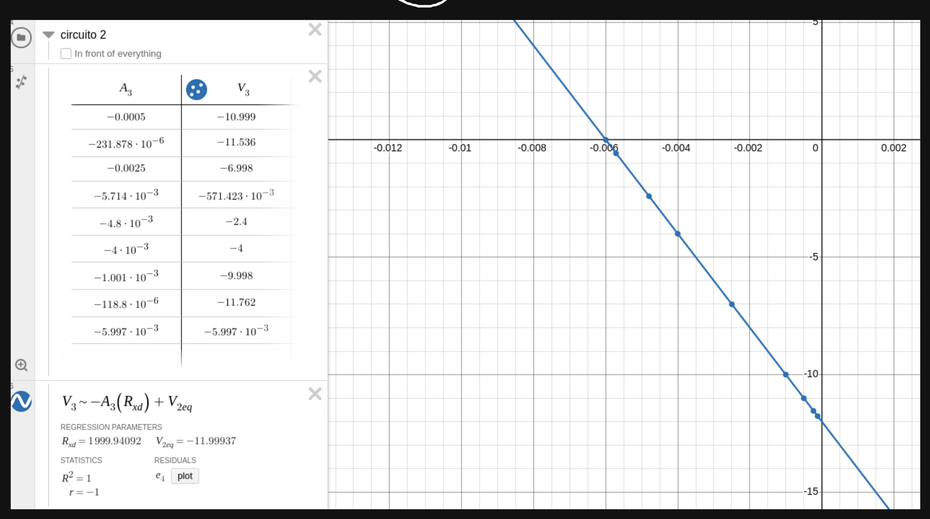
(1) $V = \frac{R_z}{R_z + R_{eq} + R_1} V_{eq}$

(2) $A = \frac{V_{eq}}{R_z + R_{eq} + R_1}$



$$V = R_z A$$

$$R_z = \frac{V_{eq}}{A} - R_{eq} - R_1 \quad \Rightarrow \quad V = V_{eq} - A(R_{eq} + R_1)$$



$R_{xd} = R_{eq} + R_1 = 2000\,\Omega$

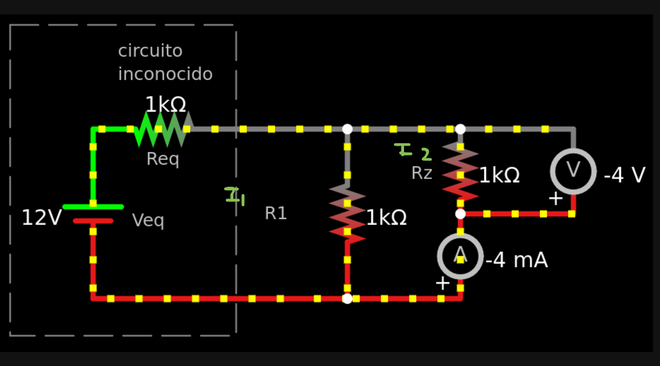
$V_{eq} \approx -12\,V$

circuito 3

$$I_T = \frac{V_{eq}}{\frac{R_z R_1}{R_z + R_1} + R_{eq}}$$

(1) $V = \frac{R_z}{\frac{R_z R_1}{R_z + R_1} + R_{eq}} V_{eq}$

(2) $A = \frac{V_{eq} - R_{eq}\left(\frac{V_{eq}}{\frac{R_z R_1}{R_z + R_1} + R_{eq}}\right)}{R_z}$



$$V = I_1 R_1 = I_2 R_2$$

$$\frac{I_1 R_1}{R_2} = I_2$$

$$\frac{R_1}{R_2}\left(\frac{V_{eq} - R_{eq} I_T}{R_1}\right)$$

